POWER PULSE

Power System Stability Basics Important Formulae

Notes by BALAJI SIR

1 Swing equation

1. $$\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e = P_a \quad \text{(pu)}, \qquad M = \frac{HS}{180f}\ \text{MJ/elec-deg}$$

2. $$H = \frac{\text{stored KE at synchronous speed (MJ)}}{S_{rated}\ \text{(MVA)}}, \qquad H_{eq} = \frac{\sum H_i S_i}{S_{base}}$$

3. $$\alpha = \frac{\pi f}{H}(P_m - P_e)\ \text{rad/s}^2$$

2 Power-angle and steady-state limit

4. $$P_e = \frac{EV}{X}\sin\delta, \qquad P_{max} = \frac{EV}{X}\ \text{at}\ \delta = 90^\circ$$

X = X'_d + X_t + X_line (transient model uses E')

5. $$P_s = \frac{dP_e}{d\delta} = P_{max}\cos\delta_0 \quad \text{(synchronising coefficient)}$$

6. $$\omega_n = \sqrt{\frac{\pi f P_s}{H}}\ \text{rad/s}, \qquad f_n = \frac{\omega_n}{2\pi}$$

small-oscillation frequency (0.5–2 Hz)

3 Time domain

7. $$\delta(t) = \delta_0 + \frac{\pi f\, P_a}{2H}t^2 \quad (P_a \text{ constant})$$

8. $$\Delta\delta_n = \Delta\delta_{n-1} + k\, P_{a,n-1}, \qquad k = \frac{180f}{H}(\Delta t)^2\ \text{(deg)}$$

point-by-point method

9. Types: steady-state, transient (large disturbance), dynamic

10. Improvement: fast clearing, auto-reclosing, series compensation, high-speed excitation